The Poisson–Dirichlet distribution is the unique invariant law for general uniform split–merge dynamics

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Abstract

We study uniform split–merge dynamics on mass partitions. At each step, two parts are sampled independently in a size-biased manner. A repeated selection triggers a uniform split with probability β_s. Two distinct selections trigger a merge with probability β_m. We prove that $\mathrm{PD}(\beta_s/\beta_m)$ is the unique invariant law in the previously unresolved regime $\beta_s > \beta_m$, completing uniqueness for all parameter values. The proof uses finite paintbox partitions and a sublinear relative-entropy estimate to derive the Ewens sampling formula.

Keywords. Split-merge dynamics, Poisson-Dirichlet, Uniqueness of invariant measure.
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1 Introduction and main result

We define

$$\Delta = \left\{ x = (x_1, x_2, \ldots) \in [0,1]^{\mathbf{N}} \,:\, \sum_{i=1}^{\infty} x_i = 1 \right\},$$

and its ordered subset $\nabla := \{p \in \Delta \,:\, p_1 \geq p_2 \geq \ldots\}$, both equipped with the ℓ^1-topology. For every nonnegative summable sequence x, $\mathrm{rank}(x)$ denotes its decreasing rearrangement (we append zeros, if x is of finite length).

We define the following discrete time Markov dynamic on ∇ for any $\beta_s, \beta_m \in (0,1]$. Given $p \in \nabla$, we size-biased sample two parts p_i and p_j of p, i.e. we choose i and j independently with probability p_i and p_j, respectively. Now independently, if $i = j$, we split the part p_i uniformly into two parts with probability β_s (and do nothing with probability $1 - \beta_s$), otherwise we merge the two parts p_i and p_j with probability β_m (and do nothing with probability $1 - \beta_m$). This Markov dynamic is more precisely represented by the split-merge kernel

$$K_{\beta_s,\beta_m}(p, \bullet) = \beta_s \sum_i p_i^2 \int_0^1 \mathbb{1}_\bullet(\mathbf{S}_{i,u} p)\, du + \beta_m \sum_{i \neq j} p_i p_j \mathbb{1}_\bullet(\mathbf{M}_{i,j} p) + \left(1 - \beta_s \sum_i p_i^2 - \beta_m \sum_{i \neq j} p_i p_j\right) \mathbb{1}_\bullet(p)\,,$$

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where the split and merge operators are defined as

$$\mathbf{S}_{i,u}p := \mathrm{rank}(p_1, \ldots, p_{i-1}, up_i, (1-u)p_i, p_{i+1}, \ldots)\,,$$
$$\mathbf{M}_{i,j}p := \mathrm{rank}(p_1, \ldots, p_{i-1}, p_i + p_j, \ldots, p_{j-1}, p_{j+1}, \ldots)\,.$$

Of course we set $\mathbf{M}_{j,i} = \mathbf{M}_{i,j}$. For every $\beta_s, \beta_m \in (0,1]$, the associated Markov chain is stationary with respect to the Poisson–Dirichlet law $\mathrm{PD}(\beta_s/\beta_m)$, see for example [MWZZ02]. The Poisson–Dirichlet distribution $\mathrm{PD}(\theta)$, $\theta > 0$, is the probability measure that can be defined through the following stick-breaking procedure [Fen10]: Let $U_1, U_2, \ldots$ be independent $\mathrm{Beta}(1,\theta)$-distributed random variables and define

$$V_1 := U_1, \qquad V_2 := (1-U_1)U_2, \qquad V_3 := (1-U_1)(1-U_2)U_3, \ldots\,.$$

In other words, starting with a stick of length one we iteratively break off random fractions proportional to U_i, and put the designated broken off piece (say the one we hold in our right hand) to the side. Subsequent decreasing reordering of the entries of $(V_i)_{i\in\mathbf{N}}$, i.e. applying rank, yields the law $\mathrm{PD}(\theta)$. The PD-law can also be realised as a limit of Dirichlet distributions.

1.1 The main result

Although $\mathrm{PD}(\beta_s/\beta_m)$ is known to be stationary for the split–merge process throughout the entire parameter range, its uniqueness as a stationary distribution remained partially open. Our main result resolves this question:

Theorem 1.1. *For every $\beta_s, \beta_m \in (0,1]$, the Poisson-Dirichlet distribution $\mathrm{PD}(\beta_s/\beta_m)$ is the unique invariant probability measure of the uniform split-merge process with kernel K_{β_s,β_m}.*

In the case $\beta_s = \beta_m$, Theorem 1.1 was proved in [DMWZZ04]. An alternative proof was later provided in [Sch05], which extends to the regime $\beta_s \leqslant \beta_m$ [GUW11, Theorem 7.1(b)]. By contrast, the regime $\beta_s > \beta_m$ remained unresolved, although the corresponding statement was widely believed to hold.

Remark 1.2. *The extension of the coupling argument in [Sch05] to unequal split and merge probabilities relies on a monotonicity estimate that holds for $\beta_s \leqslant \beta_m$. More precisely, one couples two split–merge dynamics on ∇, one starting from an invariant measure μ and the other from $\mathrm{PD}(\beta_s/\beta_m)$. The coupling tries to create matches between blocks of equal size in the two chains, possibly at different positions, and updates matched blocks together. When a split in one chain is paired with a merge in the other, the split adds at most one unmatched block and the merge removes at least one (with high probability). While for $\beta_s \leqslant \beta_m$ a split can always be coupled with a merge, such that the total count of unmatched blocks does not increase, it fails for $\beta_s > \beta_m$.*

Throughout the remainder of the paper we will restrict ourselves to the case $\beta_s > \beta_m$, for the sake of a simpler notation. Without loss of generality, we will assume that $\beta_s = 1$ and $\beta_m = \theta^{-1} < 1$. Indeed, we have

$$K_{\beta_s,\beta_m} = \beta_s K_{1,1/\theta} + (1-\beta_s)\mathrm{id}\,.$$

This proportional increase in probabilities does not affect the invariant measure. For brevity, we will write $K_{1/\theta} = K_{1,1/\theta}$ for the split-merge kernel in this regime.

1.2 Consequences of the main result

A direct application of Theorem 1.1 (and the initial motivation for this paper) concerns scaling limits of condensing interacting particle systems. More precisely, it identifies the thermodynamic limits of certain product-form stationary measures as rescaled PD(θ), by showing that these limits are invariant under split-merge dynamics [CGG22]. Thus, our theorem removes the restriction $\theta \leq 1$ in [CGG22].

Split-merge dynamics and the Poisson-Dirichlet law also appear in the study of quantum-spin systems and related random-loop models, whose configurations consist of collections of closed paths on a lattice or in space–time [GUW11, BU20]. In several such models, the macroscopic loops are conjectured to evolve effectively according to uniform split–merge dynamics, leading to Poisson–Dirichlet statistics with parameter $\theta > 1$. The relevant parameters include the Heisenberg model ($\theta = 2$) [GUW11] and spin-one models ($\theta = \frac{3}{2}, 3$) [CMUW23]. Theorem 1.1 supplies the identification of the invariant law throughout this parameter range, once every subsequential limit of the normalised loop-length distributions has been shown to be supported on ∇ and invariant under the corresponding uniform split–merge dynamics, see also [BU20, Section 4.2]. However, establishing such invariance from the underlying spatial loop model remains a separate, model-dependent problem.

Statement on the use of AI. The present paper was developed as a test case for the agentic workflow described in [Xu26]. An initial proof of Theorem 1.1 was generated by OpenAI Codex using GPT 5.6 Sol. The generated proof used a cyclic-lift representation. After reviewing the proof, the author replaced it by a partition-based reformulation which was refined through subsequent interactions with the model. Almost all of the manuscript was written in August 2026. AI assistance with the prose was limited to occasional editorial suggestions (including this statement). GPT 6 Astra was used for mathematical review and final proofreading.

The author takes full responsibility for the correctness and presentation of the paper. Materials documenting the process, including the initially generated proof, are available at `https://github.com/KugelschreiberY/PD_unique_splitmerge`.

2 Proof of the main result

2.1 Paintbox sampling

Write $\mathcal{P}_n$ for the set of partitions of $[n]$. For every $x \in \Delta$ we can associate a probability law on $\mathbf{N}$ by sampling in a size-biased fashion. Namely, we define $\mathbf{P}_x$ to be the law given in terms of

$$\mathbf{P}_x(Z = i) = x_i \,.$$

Likewise, we write $\mathbf{P}_x^{\otimes n}$ for the product measure that samples n indices $(Z_1, \ldots, Z_n)$ independently, with replacement. This is often known as paintbox sampling.

Given $(Z_1, \ldots, Z_n)$, we can define a partition Π_n of $[n]$ by grouping together those indices that sampled the same box, i.e. i and j lie in the same block of Π_n whenever

$Z_i = Z_j$. We denote the law of the resulting random partition by $\Phi_n(x, \bullet)$. The resulting partition forgets about the precise values of Z but only remembers which entries agree.

Lastly, we denote the number of distinct indices (analogously the number of blocks in the corresponding partition) by

$$R_n := |\{Z_1, \dots, Z_n\}| \,.$$

2.2 Random partitions

We say Π_n is an exchangeable random partition of $[n]$ if its law is invariant under relabelling. In particular, Π_n is exchangeable if it is generated by a sequence of iid random variables $(Z_1, \dots, Z_n)$, see for example the paintbox sampling above.

The *exchangeable partition probability function* (EPPF) of an exchangeable random partition Π_n of $[n]$ is the associated probability mass function expressed only in block sizes, i.e.

$$\mathbb{P}(\Pi_n = \{B_1, \dots, B_r\}) =: p(|B_1|, \dots, |B_r|) \,, \tag{2.1}$$

which by definition is symmetric.

Analogous to the finite setting, we can define an infinite partition Π of $\mathbf{N}$ using an exchangeable sequence of random variables $(X_i)_{i \in \mathbf{N}}$ with law μ. As in the finite case, this yields an EPPF p which is consistent when constructing Π_n on $[n]$ through $(X_1, \dots, X_n)$, see [Pit06, Section 2.2]. Notice that we can always restrict a partition Π on $\mathbf{N}$ to one on $[n]$ by removing all entries for $k > n$.

Lemma 2.1 (Finite EPPFs determine the infinite partition law)**.** *Let* Π *and* $\overline{\Pi}$ *be random partitions of* $\mathbf{N}$*. If*

$$\mathbb{P}(\Pi|_{[n]} = \pi) = \mathbb{P}(\overline{\Pi}|_{[n]} = \pi) \,, \quad \forall n \in \mathbf{N} \,,\ \pi \in \mathcal{P}_n \,,$$

then $\Pi \stackrel{d}{=} \overline{\Pi}$*.*

For example, the partition Π_{PD} generated through paintbox sampling with respect to $\mathrm{PD}(\theta)$, has EPPF

$$\mathbb{P}_{\mathrm{PD}(\theta)}(\Pi_{\mathrm{PD}}|_{[n]} = \pi) = \theta^{|\pi|} \frac{\Gamma(\theta)}{\Gamma(n+\theta)} \prod_{B \in \pi} (|B| - 1)! \qquad \forall \pi \in \mathcal{P}_n \,, \tag{2.2}$$

see for instance [Pit06, Equations (2.19)–(2.20)].

Kingman's representation theorem expresses every exchangeable random partition as a mixture of paintbox partitions. To this end, we introduce the extended space $\overline{\nabla} := \{p \in [0,1]^{\mathbb{N}} : p_1 \geq p_2 \geq \cdots ,\ \sum_i p_i \leq 1\}$, with the product topology. Whenever sampling from the remaining mass $1 - \sum_i p_i$, a separate singleton block is formed.

Lemma 2.2 (Kingman's representation)**.** *Let* Π *be a random exchangeable partition of* $\mathbf{N}$*. Then there exists a unique law* $\mu \in \mathcal{M}_1(\overline{\nabla})$ *such that* Π *is the partition generated through paintbox sampling with respect to* μ*.*

This is Kingman's representation theorem [Kin78], see also [Pit06, Theorem 2.2].

2.3 Proof of Theorem 1.1

If μ is a stationary measure of the split-merge process, then it suffices to prove

$$\mathbb{P}(\Pi_\mu|_{[n]} = \pi) = \theta^{|\pi|}\frac{\Gamma(\theta)}{\Gamma(n+\theta)}\prod_{B\in\pi}(|B|-1)! \qquad \forall \pi \in \mathcal{P}_n\,, \tag{2.3}$$

to identify it as the Poisson–Dirichlet law $\mathrm{PD}(\theta)$, as a consequence of Lemmas 2.1 and 2.2. The relation (2.3) is the content of Corollary 3.6. Throughout the remainder of the paper, denote by ν_n the law of $\Pi_\mu|_{[n]}$. Analogously, $\nu_n := \mu\Phi_n$ where Φ_n was defined in Section 2.1.

Proof of Theorem 1.1. We restrict ourselves to the case $\theta > 1$, since $\theta \in (0,1]$ has been treated already in [GUW11, Theorem 7.1(b)] separately. Moreover, it suffices to establish uniqueness of invariant measures on ∇, since invariance (in fact reversibility) is established in [MWZZ02, Theorem 3(a)].

Let μ be an invariant law with respect to $K_{1,1/\theta}$. Furthermore, denote by Π the random partition of $\mathbf{N}$ induced by size-biased sampling under μ, namely $k, \ell \in \mathbf{N}$ lie in the same subset provided $Z_k = Z_\ell$. Likewise, ν_n induces a random partition Π_n of $[n]$ and by definition of ν_n through Φ_n, we have $\Pi_n = \Pi|_{[n]}$.

Corollary 3.6 yields (2.3) for every $n \in \mathbf{N}$ and $\pi \in \mathcal{P}_n$. Thus, Lemma 2.1 implies that the law of Π must agree with the law of Π_{PD}. Lastly, Kingman's representation (Lemma 2.2) yields a one-to-one correspondence between laws of exchangeable partitions and laws on $\overline{\nabla}$ via paintbox sampling. This concludes $\mu = \mathrm{PD}(\theta)$. □

3 Identification of the finite partition laws

To prove Corollary 3.6, we lift the split–merge dynamics to finite set partitions and establish its relation to the original process through paintbox sampling. Coherence and a sublinear relative entropy estimate then allow us to transfer stationarity of the original process to the finite partition laws ν_n. Finally, the resulting balance relations identify ν_n with the Ewens measure η_n.

3.1 Boundary intertwining

Let $a, b \in [n]$ be distinct. We define a Markov kernel $Q_n^{a,b}$ on $\mathcal{P}_n$. Fix $\pi \in \mathcal{P}_n$, then

- If a, b lie in distinct blocks in π, then $Q_n^{a,b}$ joins the two blocks with probability θ^{-1} and leaves π unchanged with probability $1 - \theta^{-1}$.
- If a, b lie in the same block C in π, then the block is split according to the following rule: for every partition $A \sqcup B = C \setminus \{a, b\}$ the block C is replaced by the blocks $\{a\} \cup A$ and $\{b\} \cup B$ with probability

$$\int_0^1 u^{|A|}(1-u)^{|B|}\,\mathrm{d}u = \frac{|A|!|B|!}{(|C|-1)!}\,. \tag{3.1}$$

  That is, one first samples a uniform random variable U and then assigns each element to A with probability U, independently conditional on U.

Let $n \in \mathbf{N}$ and denote by $R_A : \mathcal{P}_n \to \mathcal{P}_{n-|A|}$ the operator that removes all labels in A and relabels the remaining ones in increasing order. The following lemma shows that the family of measures $(\nu_n)_n$ is coherent in the sense that R_A acts as a projection, even after application of Q.

Lemma 3.1 (Coherence)**.** *Let $n \geqslant 2$ and $A \subset [n]$, then*

(i) $(R_A)_* \nu_n = \nu_{n-|A|}$

(ii) $(R_A)_*(\nu_n Q_n^{a,b}) = \nu_{n-|A|} Q_{n-|A|}^{a,b}$ *for all distinct $a, b < \min A$.*

Proof. The statement in (i) is immediate, since removing paintbox samples does not affect other paintboxes due to independent sampling.

For statement (ii) we notice that the merging of blocks is unaffected by removal of labels A. Likewise when splitting, each label is associated to the a-block or b-block independently (conditional on U) and their later removal has no effect. □

The following result relates the split-merge dynamic to the dynamic on partitions driven by $Q_n^{a,b}$. This requires two arbitrary drivers a, b that take the role of sampling the split-merge action in $K_{1/\theta}$. This explains the discrepancy in degrees of Φ in the statement below.

Proposition 3.2 (Split–merge intertwining)**.** *Let $n \geqslant 2$. For every probability measure μ on ∇, and distinct $a, b \in [n]$*

$$(R_{a,b})_* \mu \Phi_n Q_n^{a,b} = \mu K_{1/\theta} \Phi_{n-2} \,.$$

Proof. It suffices to prove for every $p \in \nabla$

$$\sum_{\substack{\pi' \in \mathcal{P}_n \\ R_{a,b}\pi' = \sigma}} \sum_{\pi \in \mathcal{P}_n} \Phi_n(p, \pi) Q_n^{a,b}(\pi, \pi') = \int K_{1/\theta}(p, dp') \Phi_{n-2}(p', \sigma) \,, \quad \forall \sigma \in \mathcal{P}_{n-2} \,.$$

Let $(I, J) \sim \mathbf{P}_p^{\otimes 2}$, which will play two roles: for the left-hand side it will denote the marginal distribution of $(Z_1, \ldots, Z_n) \sim \mathbf{P}_p^{\otimes n}$ for a, b, whereas for the right-hand side it will denote the choice of the two blocks the split-merge is acting on. For convenience, we condition on the event $(I, J) = (i, j)$ which occurs with probability $p_i p_j$, as well as on an independent $\mathrm{Unif}[0, 1]$ random variable $U = u$. Next we compare both the left and the right–hand side and prove the identity on each splitting/merging event.

- If $i = j$, then (on the left-hand side) Q will split the block C that both a and b lie in. Now assign each element in $C \setminus \{a, b\}$ the label i_a with probability u, and the label i_b with probability $(1 - u)$. The elements labelled i_a will be grouped with a, whereas the elements labelled i_b will be grouped with b. We denote the resulting partition by Π'.

  Now, the law $\Phi_{n-2}(\mathrm{S}_{i,u} p, \cdot)$ is the same as $\Phi_{n-2}(p, \cdot)$ with the only difference that if the paintbox samples i, it will be instead given label i_a with probability u and label i_b with probability $(1 - u)$, cf. (3.1). This yields

  $$\mathbb{P}\big(R_{a,b}\Pi' = \sigma \mid I = J = i, U = u\big) = \Phi_{n-2}(\mathrm{S}_{i,u} p, \sigma) \,.$$

To see that deletion by $R_{a,b}$ does not destroy this relation, we realise the paintbox leading to $\Phi_{n-2}(\mathrm{S}_{i,u}p)$ and insert a, b back into $[n-2]$ (before forgetting their exact labels) with fixed labels i_a and i_b. This extends $\Phi_{n-2}(\mathrm{S}_{i,u}p, \cdot)$ to a random partition in $\mathcal{P}_n$, which agrees with Π'.

Averaging over $U = u$, will yield the required integral present in both split operations.

- Let $V \sim \mathrm{Bernoulli}(\theta^{-1})$ be independent of all samples and U. If $i \neq j$, then $Q_n^{a,b}$ joins the two blocks that a, b lie in with probability $\mathbb{P}(V = 1) = \theta^{-1}$. Again it will be convenient to condition on $V = 1$. Likewise $K_{1/\theta}(p, \bullet)$ will merge p_i and p_j provided $V = 1$, such that

$$\mathbb{P}\big(R_{a,b}\Pi' = \sigma \mid I = i, J = j, V = 1\big) = \Phi_{n-2}(\mathrm{M}_{i,j}p, \sigma)\,.$$

On the other hand, for $V = 0$ (with probability $1-\theta^{-1}$) the partition is unchanged and after deleting a, b, its conditional law is $\Phi_{n-2}(p, \cdot)$.

This concludes the proof. □

Corollary 3.3. *Let μ be an invariant probability measure with respect to $K_{1/\theta}$ and write $\nu_n = \mu\Phi_n$. Then*

$$(R_{a,b})_*\nu_n Q_n^{a,b} = \nu_{n-2}\,, \quad \forall n \geqslant 2\,.$$

3.2 Control of the relative entropy

We define the Ewens measure

$$\eta_n(\pi) = \frac{\theta^{|\pi|}}{(\theta)_n} \prod_{B \in \pi} (|B| - 1)!\,, \tag{3.2}$$

where $(\theta)_n := \Gamma(n + \theta)/\Gamma(\theta)$. By definition this family of measures is coherent, i.e. $(R_A)_*\eta_n = \eta_{n-|A|}$ since $\eta_n = \mathrm{PD}(\theta)\Phi_n$ by (2.2). Moreover, the intertwining kernel $Q_n^{a,b}$ is η_n-reversible: consider a split $\pi \mapsto \pi'$ that replaces a block C containing a, b by $C_1 \ni a$ and $C_2 \ni b$, where $C = C_1 \sqcup C_2$. Hence, (3.2), (3.1), and the merge probability θ^{-1} yield

$$\frac{\eta_n(\pi')}{\eta_n(\pi)} = \theta \frac{(|C_1| - 1)!(|C_2| - 1)!}{(|C| - 1)!} = \frac{Q_n^{a,b}(\pi, \pi')}{Q_n^{a,b}(\pi', \pi)}\,.$$

Lemma 3.4 (Sublinear entropy)**.** *For every $n \geqslant 2$, we have*

$$H(\nu_n \| \eta_n) \lesssim_\theta (\log n)^2\,.$$

Proof. To every partition $\pi \in \mathcal{P}_n$ we associate a block profile $a = (a_1, \ldots, a_n)$ where $a_k := \#\{B \in \pi : |B| = k\}$. For a random partition, we denote its (random) block profile by $A = (A_1, \ldots, A_n)$.

First, we notice that ν_n only depends on the block profile of a partition, cf. (2.1). By definition the same is true for η_n. Thus, conditioning on a block profile a, we have $\nu_n(\cdot|a) = \eta_n(\cdot|a)$ which are uniform distributions amongst the partitions

$$\frac{n!}{\prod_k (k!)^{a_k} a_k!} =: N(a)$$

satisfying the given block profile a. Therefore, denoting by $\overline{\nu}_n$ the law of A under ν_n and similarly for $\overline{\eta}_n$, we have the explicit formula

$$H(\nu_n \| \eta_n) = \sum_a \sum_{\pi \leftrightarrow a} \nu_n(\pi) \log \frac{\nu_n(\pi)}{\eta_n(\pi)} = \sum_a \sum_{\pi \leftrightarrow a} \frac{\overline{\nu}_n(a)}{N(a)} \log \frac{\overline{\nu}_n(a)}{\overline{\eta}_n(a)} = H(\overline{\nu}_n \| \overline{\eta}_n)\,.$$

For $\overline{\eta}_n$ we have the explicit formula

$$\overline{\eta}_n(a) = \sum_{\pi \in \mathcal{P}_n} \eta_n(\pi) \mathbb{1}_{\pi \text{ has profile } a} = \theta^{\sum_k a_k} \frac{n!}{(\theta)_n} \prod_k \frac{1}{k^{a_k} a_k!}\,.$$

Then using the fact that $\mathrm{Ent}(\overline{\nu}_n) \geqslant 0$ yields

$$\begin{aligned} H(\overline{\nu}_n \| \overline{\eta}_n) &\leqslant -\sum_a \overline{\nu}_n(a) \log \overline{\eta}_n(a) = \mathbb{E}_{\nu_n}[-\log \overline{\eta}_n(A)] \\ &= -\mathbb{E}_{\nu_n}[R_n] \log \theta + \log \frac{(\theta)_n}{n!} + \mathbb{E}_{\nu_n}\left[\sum_k A_k \log k\right] + \mathbb{E}_{\nu_n}\left[\log\Big(\prod_k A_k!\Big)\right], \end{aligned}$$

where we recall that $\sum_k A_k = R_n$. Notice that the first term on the right-hand side can be dropped, since $\theta > 1$, and the second term yields $\log\left((\theta)_n/n!\right) \lesssim_\theta \log n$. The remaining two terms can be bounded as follows:

$$\begin{aligned} \mathbb{E}_{\nu_n}\left[\sum_k A_k \log k\right] + \mathbb{E}_{\nu_n}\left[\log\Big(\prod_k A_k!\Big)\right] &\leqslant \mathbb{E}_{\nu_n}[R_n] \log n + \mathbb{E}_{\nu_n}\left[\log\left(n^{R_n}\right)\right] \\ &\leqslant 2\mathbb{E}_{\nu_n}[R_n] \log n \lesssim_\theta (\log n)^2\,, \end{aligned}$$

where the final step is a consequence of Corollary A.3. □

3.3 Identification of the Ewens sampling formula

Lemma 3.5. *For every $n \in \mathbf{N}$ and distinct $a, b \in [n]$, we have $\nu_n Q_n^{a,b} = \nu_n$.*

The proof of this lemma will be stated at the end of this section. First, we conclude that invariance of ν_n with respect to $Q_n^{a,b}$ immediately yields identification of ν_n as the Ewens sampling formula.

Corollary 3.6. *For every n we have $\nu_n = \eta_n$. In other words,*

$$\mathbb{P}_\mu(\Pi|_{[n]} = \pi) = \frac{\theta^{|\pi|}}{(\theta)_n} \prod_{B \in \pi} (|B| - 1)!\,, \quad \forall \pi \in \mathcal{P}_n\,. \tag{3.3}$$

Proof. Let $\pi \in \mathcal{P}_n$ and fix distinct $a, b \in [n]$ such that they lie in the same block C (assuming such a block exists). Moreover, define π' to be the partition generated from π by splitting off the singleton $\{a\}$ from C, i.e. replacing C by $\{a\}$ and $C \setminus \{a\}$. By Lemma 3.5, we have

$$\begin{aligned}\nu_n(\pi') &= \nu_n Q_n^{a,b}(\pi') = \nu_n(\pi) Q_n^{a,b}(\pi, \pi') + \nu_n(\pi') Q_n^{a,b}(\pi', \pi') \\ &= \frac{(|C|-2)!}{(|C|-1)!} \nu_n(\pi) + (1 - \theta^{-1}) \nu_n(\pi') \,,\end{aligned}$$

which implies $\nu_n(\pi) = \theta^{-1}(|C|-1)\nu_n(\pi')$. Consequently, we can decompose the blocks of any partition π into singletons, i.e. the identity, by iteratively applying Q_n with suitable choices of a, b. It will take $n - |\pi|$ such splittings, such that

$$\nu_n(\pi) = \nu_n(\mathrm{id}) \theta^{|\pi|-n} \prod_{C \in \pi} (|C|-1)! \,.$$

On the other hand, unit total mass requires

$$\theta^n \nu_n(\mathrm{id})^{-1} = \sum_{\pi \in \mathcal{P}_n} \theta^{|\pi|} \prod_{C \in \pi} (|C|-1)! = (\theta)_n \,,$$

where the last equality follows from normalization of η_n, see (2.2). Hence, the probability mass function of ν_n agrees with the Ewens distribution η_n. The proof is complete after noting that $\Pi|_{[n]} \sim \nu_n$. □

Before moving to the proof of Lemma 3.5, we notice the following useful fact:

Lemma 3.7. *Let Ω be a finite set and Q a Markov kernel that is η-reversible. Then for every probability law $\nu \ll \eta$ on Ω, we have*

$$H(\nu Q \| \eta) = H(\nu \| \eta) \quad \Rightarrow \quad \nu Q^2 = \nu \,.$$

Proof. The proof is a simple consequence of the inequality

$$H(\nu \| \nu Q^2) \leqslant H(\nu \| \eta) - H(\nu Q \| \eta) \,,$$

see for example [HH09, Theorem 6]. □

Proof of Lemma 3.5. We split the proof into smaller blocks.

Step 1. Let $n \in \mathbf{N}$ and $a, b \in [n]$, and let us abbreviate $Q_n = Q_n^{a,b}$. If we know that

$$H(\nu_n Q_n \| \eta_n) = H(\nu_n \| \eta_n) \,, \tag{3.4}$$

then Lemma 3.7 will imply that $\nu_n Q_n^2 = \nu_n$. The desired invariance with respect to Q_n holds if we establish that the kernel is aperiodic. Indeed, fix any partition $\pi \in \mathcal{P}_n$ and denote by $C_1, \dots C_r \in \pi$ all blocks that neither contain a nor b. The kernel $Q_n = Q_n^{a,b}$ only gives positive mass to partitions with the given block structure C_i, i.e. that lie in

$$\big\{ \pi' \in \mathcal{P}_n : \{ C' \in \pi' : C' \cap \{a, b\} = \varnothing \} = \{C_1, \dots, C_r\} \big\} \,,$$

on which it acts irreducibly. In every such set there is at least one partition π' in which a, b lie in distinct blocks such that $Q_n(\pi', \pi') = 1 - \theta^{-1} > 0$. Thus, Q_n is aperiodic which lifts invariance with respect to Q_n^2 to $\nu_n Q_n = \nu_n$. It is left to prove (3.4), which we do in the remaining steps.

Step 2. Next, we define

$$\delta_n := H(\nu_n \| \eta_n) - H(\nu_n Q_n \| \eta_n) \geqslant 0\,,$$

and establish that δ_n is increasing in n.

For $h : \mathcal{P}_n \to \mathbf{R}$ we write $Q_n h(\pi) = \sum_{\pi'} Q_n(\pi, \pi') h(\pi')$ for the semigroup action. Throughout the proof, we will use the following property of relative entropy: if Q is a Markov kernel, then

$$H(\nu Q \| \eta Q) \leqslant H(\nu \| \eta)\,. \tag{3.5}$$

Moreover, we define $f_{n,m} : \mathcal{P}_m \to \mathbf{R}$ with

$$f_{n,m} := \frac{\nu_n}{\eta_n} \circ R_n\,,$$

where we wrote $R_n := R_{n+1:m}$ for the restriction of partitions to the subset $[n]$, and write f_m for $f_{m,m}$. Following the proof of Lemma 3.1, we have

$$Q_m(h \circ R_n) = (Q_n h) \circ R_n\,, \tag{3.6}$$

for $m \geq n$ and every $h : \mathcal{P}_n \to \mathbb{R}$. Moreover, reversibility implies

$$\nu_n Q_n = (f_n \eta_n) Q_n = (Q_n f_n) \eta_n \tag{3.7}$$

which, together with (3.6), gives

$$Q_m f_{n,m} = Q_n f_n \circ R_n = \frac{\nu_n Q_n}{\eta_n} \circ R_n\,. \tag{3.8}$$

Now, we show that

$$H(\nu_m Q_m \| \eta_m) = H(\nu_n Q_n \| \eta_n) + H(\nu_m Q_m \| (Q_m f_{n,m}) \eta_m)\,. \tag{3.9}$$

To this end, we use the decomposition

$$\log \frac{\nu_m Q_m}{\eta_m} = \log Q_m f_{n,m} + \log \frac{\nu_m Q_m}{(Q_m f_{n,m}) \eta_m}\,. \tag{3.10}$$

The last term on the right-hand side precisely generates the last term in (3.9). Therefore, it is only left to recover $H(\nu_n Q_n \| \eta_n)$: by application of (3.8) we see that

$$\begin{aligned}\sum_{\pi \in \mathcal{P}_m} \nu_m Q_m(\pi) \log (Q_m f_{n,m})(\pi) &= \mathbb{E}_{\nu_m Q_m}\left[\log \frac{\nu_n Q_n \circ R_n}{\eta_n \circ R_n}\right] \\ &= \mathbb{E}_{\nu_n Q_n}\left[\log \frac{\nu_n Q_n}{\eta_n}\right] = H(\nu_n Q_n \| \eta_n)\end{aligned}$$

where in the second equality we applied Lemma 3.1(ii).

Likewise, we have

$$H(\nu_m\|\eta_m) = H(\nu_n\|\eta_n) + H(\nu_m\|f_{n,m}\eta_m)\,, \tag{3.11}$$

which follows the same decomposition as in (3.10) but by inserting $f_{n,m}$, and using Lemma 3.1(i) thereafter. We subtract (3.9) from (3.11) and furthermore apply the bound,

$$H(\nu_m Q_m\|(Q_m f_{n,m})\eta_m) = H(\nu_m Q_m\|(f_{n,m}\eta_m)Q_m) \leqslant H(\nu_m\|f_{n,m}\eta_m)\,, \tag{3.12}$$

which is a consequence of (3.7) and (3.5). This establishes $\delta_n \leqslant \delta_m$.

Step 3. Using Corollary 3.3 we have

$$\begin{aligned} H(\nu_m Q_m\|\eta_m) &\geqslant H\big((R_{a,b})_*\nu_m Q_m\|(R_{a,b})_*\eta_m\big) \\ &= H\big((R_{a,b})_*\nu_m\|(R_{a,b})_*\eta_m\big) = H(\nu_{m-2}\|\eta_{m-2})\,, \end{aligned}$$

where the first inequality follows from (3.5), and the second equality is a consequence of coherence, see Lemma 3.1. Overall, this establishes that for every $m \geq n+2$

$$\delta_n \leqslant \delta_m \leqslant H(\nu_m\|\eta_m) - H(\nu_{m-2}\|\eta_{m-2})\,. \tag{3.13}$$

Lastly, telescoping and applying (3.13) yields

$$m\delta_n \leqslant H(\nu_{n+2m}\|\eta_{n+2m}) - H(\nu_n\|\eta_n) \lesssim_\theta (\log(n+2m))^2\,,$$

where the last inequality follows from Lemma 3.4. Dividing both sides by m and taking the limit $m \to \infty$ yields (3.4). □

A Moment estimates

First, we recall the following standard identity, see [MWZZ02, Lemma 5].

Lemma A.1. *Let μ be an invariant probability measure of the split-merge process with kernel $K_{1/\theta}$. Then*

$$\mathbb{E}_\mu\Big[\sum_i p_i^2\Big] = \frac{1}{\theta+1}\,.$$

Next, we define

$$U_n := \mathbb{E}_\mu\Big[\sum_i p_i(1-p_i)^n\Big] \quad \text{and} \quad V_n := \mathbb{E}_\mu\Big[\sum_{i\neq j} p_i p_j(1-p_i-p_j)^n\Big]\,.$$

In other words, U_n is the (annealed) probability that we size-biasedly picked p_i after having drawn n-times a different block before. The following lemma shows that this probability vanishes linearly.

Lemma A.2. *Let μ be an invariant probability measure of the split-merge process with kernel $K_{1/\theta}$. Then*

$$U_n \leq \frac{2\theta}{n+1}\,.$$

Proof. Let μ be an invariant probability measure. We claim the identity

$$U_n - U_{n+1} + \frac{2}{\theta}U_{n+1} - \frac{1}{\theta}V_n = \frac{2}{n+1}(1-U_{n+1}) \tag{A.1}$$

which implies the statement of the lemma, because $0 \leqslant U_{n+1} \leqslant U_n$ by definition, and

$$V_n = \mathbb{E}_\mu\Big[\sum_{i\neq j} p_i p_j (1-p_i-p_j)^n\Big] \leqslant \mathbb{E}_\mu\Big[\sum_i p_i(1-p_i)^n \sum_{j\neq i} p_j\Big] = U_{n+1}\,.$$

In order to prove (A.1), consider functions

$$F_n(p) := \sum_i \big(1-(1-p_i)^n\big)\,,$$

with $F_0 \equiv 0$ and perform a single split-merge step. Stationarity yields $\mathbb{E}_\mu[K_{1/\theta}F_n - F_n] = 0$ and therefore yields a balance of splitting and merging when acting on F_n. More precisely, the splitting-contribution of $\mathbb{E}_\mu[K_{1/\theta}F_n - F_n]$ is given in terms of

$$\begin{aligned}
&\mathbb{E}_\mu\Big[\sum_i p_i^2\Big(\int_0^1 F_n(\mathbf{S}_{i,u}p)\,du - F_n(p)\Big)\Big] \\
&= \mathbb{E}_\mu\Big[\sum_i p_i^2\Big(\int_0^1 1-(1-up_i)^n-(1-(1-u)p_i)^n\,\mathrm{d}u + (1-p_i)^n\Big)\Big] \\
&= \frac{1}{1+\theta} - 2\frac{1}{(n+1)}\Big(1-U_{n+1}\Big) + U_n - U_{n+1}\,.
\end{aligned}$$

where in the last step we applied Lemma A.1, and used

$$\int_0^1 (1-up_i)^n\,du = \frac{1-(1-p_i)^{n+1}}{(n+1)p_i}\,, \quad \text{provided } p_i > 0\,,$$

as well as $\mathbb{E}_\mu\big[\sum_i p_i^2(1-p_i)^n\big] = U_n - U_{n+1}$.

Likewise, the merging-contribution of $\mathbb{E}_\mu[K_{1/\theta}F_n - F_n]$ is given in terms of

$$\begin{aligned}
&\frac{1}{\theta}\mathbb{E}_\mu\Big[\sum_{i\neq j} p_i p_j\Big(\int_0^1 F_n(\mathbf{M}_{i,j}p)\,du - F_n(p)\Big)\Big] \\
&= \frac{1}{\theta}\mathbb{E}_\mu\Big[\sum_{i\neq j} p_i p_j\Big((1-p_i)^n + (1-p_j)^n - (1-(p_i+p_j))^n - 1\Big)\Big] \\
&= \frac{2}{\theta}U_{n+1} - \frac{1}{\theta}V_n - \frac{1}{1+\theta}\,.
\end{aligned}$$

Summing both merging and splitting contributions yields (A.1), since the constant terms cancel precisely. □

Equipped with Lemma A.2, we can show that the expected number of distinct blocks R_n in a paintbox sample of size n grows at most logarithmically when sampling with respect to an invariant split-merge measure.

Corollary A.3. *Let μ be an invariant probability measure of the split-merge process with kernel $K_{1/\theta}$. Then*

$$\mathbb{E}_\mu[R_n] \lesssim_\theta \log n\,.$$

Proof. First, we notice that $R_{n+1} - R_n = \mathbb{1}_{Z_{n+1}\notin\{Z_1,\dots,Z_n\}}$, which implies

$$\mathbb{E}_\mu[R_{n+1} - R_n] = \mathbb{P}_\mu\big(Z_{n+1} \notin \{Z_1,\dots,Z_n\}\big) = \mathbb{E}_\mu\bigg[\sum_i (1-p_i)^n p_i\bigg] = U_n\,.$$

Consequently, $\mathbb{E}_\mu[R_n] = 1 + \sum_{m=1}^{n-1} U_m$. The statement follows from Lemma A.2 because $\sum_{m=1}^{n} m^{-1} \sim \log n$. □

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